

ECO 362 Financial Investments

Precept 10: Options / Taxes

December 1 – December 4, 2025

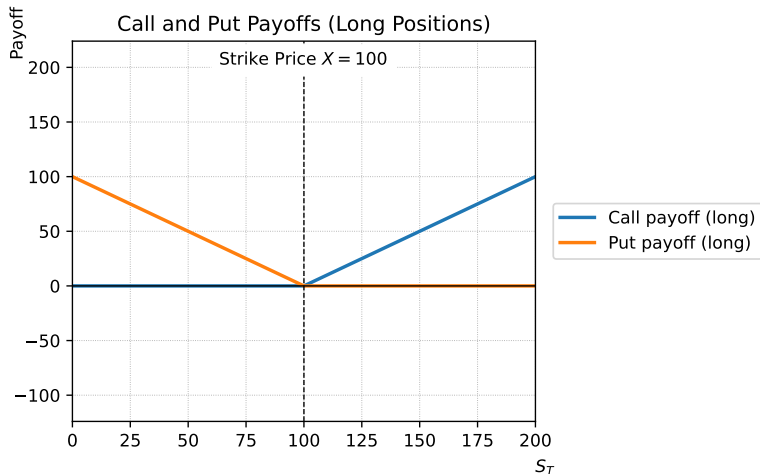
1 Options

2 Taxes

*Note: Slides marked with $*_{new}$ contain additional graphical explanations compared to 2024.*

- Terms of the (European) option contracts
 - Some underlying asset whose market price is S_t at any time t
 - Price to pay today: C for call option or P for put option
 - Time to expiration T
 - Exercise price (or strike price) at expiration date X
- How does (call) option work?
 - Buyer pays seller option price C **today**;
 - Buyer has the **right** to buy an asset at price X at the maturity date T .
- Payoff functions
 - Call option buyer: $\max(S_T - X, 0)$
 - Put option buyer: $\max(X - S_T, 0)$

where S_t is the price of the underlying asset at time t .



- Relation between prices of a call and a put with the same terms (S, T, X)

$$C = P + S_0 - Xe^{-r_b T}$$

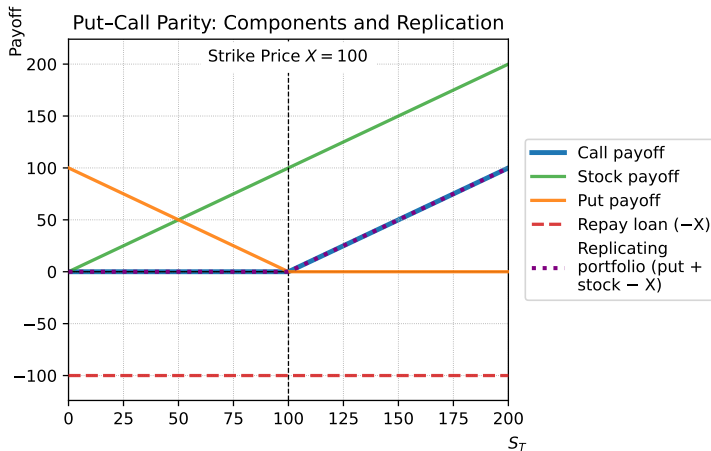
- Derivation from no arbitrage argument

- Trading strategy (to replicate a call option):

- Pay P to long a put with exercise price X and time to expiration T
- Buy the asset at price S_0
- Borrow $Xe^{-r_b T}$ at riskless interest rate

- Payoffs from the strategy

- Total cost now: $P + S_0 - Xe^{-r_b T}$
- Total payoff at expiration T : $\max(X - S_T, 0) + S_T - X = \max(S_T - X, 0)$
- Same as a call with exercise price X and time to expiration T
- By no arbitrage, the cost should be the same $C = P + S_0 - Xe^{-r_b T}$



Options: Black-Scholes Formula

- Black-Scholes specifies a theoretical option price given (S_0, T, X, σ)
- Call price is

$$C = S_0 \mathbb{N}(d_1) - Xe^{-r_b T} \mathbb{N}(d_2)$$

where

$$d_1 = \frac{\log(S_0/X) + (r_b + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

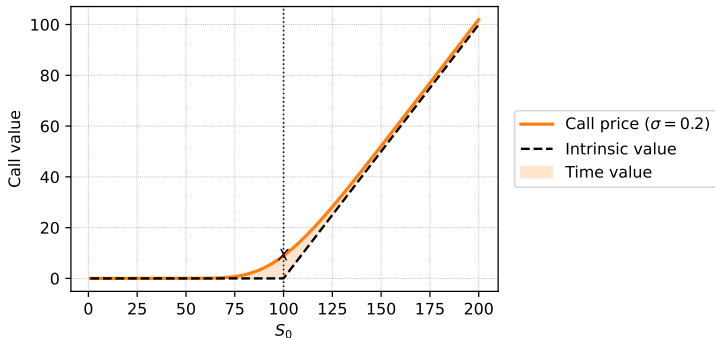
$\mathbb{N}(\cdot)$ is the CDF of the standard normal distribution (mean 0, standard deviation 1)

- By put-call parity, the put price is

$$P = -S_0(1 - \mathbb{N}(d_1)) + Xe^{-r_b T}(1 - \mathbb{N}(d_2))$$

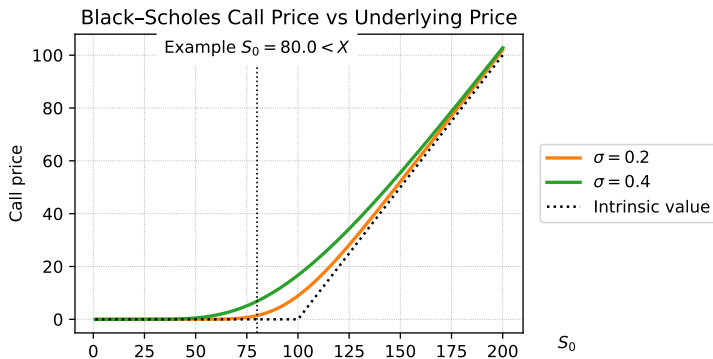
- Call price can be decomposed into two parts:

$$C = \underbrace{\max(S_0 - X, 0)}_{\text{intrinsic value}} + \underbrace{(C - \max(S_0 - X, 0))}_{\text{time value}}$$

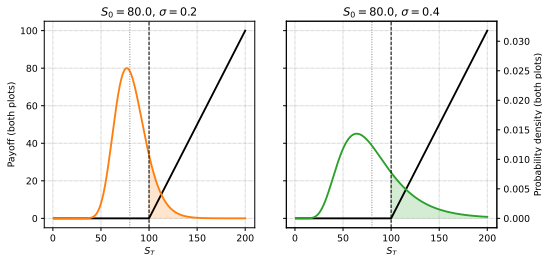


- Intrinsic value: immediate exercise value if we could exercise now
- Time value: value from potential future price movements above strike X

- For a call option with $S_0 < X$, intrinsic value is zero, so all value comes from time value and:
 - More time to expiration $T \Rightarrow$ more opportunity for S_T to exceed X
 - Higher volatility $\sigma \Rightarrow$ more spread out possible S_T values
- i.e. for out-of-the-money options, option value reflects the possibility that S_T exceeds X at maturity.



- Think of the future price S_T as distributed over many possible outcomes.
 - Some outcomes have $S_T < X \Rightarrow$ payoff is zero
 - Some outcomes have $S_T > X \Rightarrow$ payoff is positive, $S_T - X$
- The option price today reflects a probability-weighted average of outcomes where $S_T > X$. Finding the related "risk-neutral" probability distribution is beyond the scope of this course. However, intuitively:
- Higher volatility makes the distribution of S_T wider, increasing the chance of $S_T > X$ outcomes.



Options: Implied Volatility

- Definition: Suppose the Black-Scholes model is right, what value of σ is consistent with market price?
- $\Rightarrow$ Implied volatility σ_{imp} is the value of σ that, when plugged into Black-Scholes formula, gives the market price of the option that is actually observed. Use a solver for this.
- Why might this be useful?
 - 1 A different way to quote option price; similar to bond price and yield.
 - Higher implied volatility is equivalent to higher price, everything else equal
 - An option is more valuable because of its exposure to downside or upside risk
 - 2 To test Black-Scholes model
 - Observe multiple options with the same underlying asset in the markets
 - Black-Scholes model requires they have the same value for volatility σ
 - Compute implied volatilities for these options
 - A volatility smile is evidence against Black-Scholes model

1 Options

2 Taxes

*Note: Slides marked with $*_{new}$ contain additional graphical explanations compared to 2024.*

- Let τ_i be the income tax rate and τ_c be the capital gains tax rate
- The return before tax is

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1 = \underbrace{\frac{P_t}{P_{t-1}} - 1}_{\text{capital gain}} + \underbrace{\frac{D_t}{P_{t-1}}}_{\text{dividend yield}}$$

- After-tax return if capital gain is unrealized

$$R'_t = \frac{P_t}{P_{t-1}} - 1 + \frac{(1 - \tau_i)D_t}{P_{t-1}}$$

- After-tax return if capital gain is realized

$$R'_t = (1 - \tau_c)\left(\frac{P_t}{P_{t-1}} - 1\right) + \frac{(1 - \tau_i)D_t}{P_{t-1}}$$

Taxes: Accounts with Tax Advantages

- Traditional 401(k):
- Earnings are tax-deferred until withdrawal at time T :

$$(1 - \tau_{i,T}) \prod_{t=1}^T (1 + R_t)$$

- Roth 401(k):
- Payments into Roth are taxed, but earnings are tax-free upon withdrawal at time T :

$$(1 - \tau_{i,0}) \prod_{t=1}^T (1 + R_t)$$

Choice between the two depends on expected tax rates at contribution and withdrawal times.